Leetcode 91. Decode Ways

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Understanding the Problem

Given an encoded message where each number or pair of numbers represents a letter in the alphabet. For example, "1" $\rightarrow$ 'A' and "20" $\rightarrow$ 'T'. This, however, means that a message can be decoded in multiple ways since two numbers can either be separated or combined. For example, "11" $\rightarrow$ "AA", "K".

Solution

This problem can be solved with **dynamic programming** since we can use smaller substrings and gradually combine the intermediate results to get the final answer. However, before starting, it's important to address the many cases for this problem and what to do for each case.

I Two consecutive "0"

If the input contains two consecutive "0" characters, then return 0 since it cannot be decoded. Example: "19581958$\underline{00}$8518".

II All characters are either "1" or "2"

In this case, all characters can be combined without creating any invalid combos. Considering small examples is useful here:

$$1 \rightarrow \{1\} = 1$$

$$11 \rightarrow \{(1,1),(11)\} = 2$$

$$111 \rightarrow \{(1,1,1),(11,1),(1,11)\} = 3$$

$$1111 \rightarrow \{(1,1,1,1),(11,1,1),(1,11,1),(1,1,11),(11,11)\} = 5$$

Notice how the outputs appear to resemble the Fibonacci sequence. This is because when a number is added, it can be combined or left separate (example:

1111 and 1111). This creates two distinct subproblems that can be combined. Therefore, we get the following recurrence relation:

$$F(n) = F(n-1) + F(n-2)$$

where n is the length of the string. In the example above, $F(4) = F(3) + F(2) = 3 + 2 = 5$.

III Handling "0"

When there is a "0" in the input, it reduces the number of valid combinations that could be made. Take the following examples: "111" and "110". We know from II that "111" $\rightarrow$ 3. However, note that "110" $\rightarrow$ 1. This is because "0" must be combined with "1", leading to only one decoding: (1, 10). This means that only one subproblem can be derived, $F(n-2)$, since $F(n-1)$ leads to "110", an invalid combination:

$$F(n) = F(n-2)$$

IV Handling other characters

Having any character larger than "2" can also create invalid combinations. For example, consider "229". There are only 2 decodings, $\{(2, 2, 9), (22, 9)\}$, since "29" is not a valid character. This is because only one subproblem can be derived, $F(n-1)$, since "9" and "2" must remain separate. So, in this case:

$$F(n) = F(n-1)$$

There is also a case where both characters cannot be combined or left separate. For example, "39". This case is treated exactly as in I, i.e. 0 should be returned. There can also be a valid combination of 2 characters that use "1" or "2" and other characters. If both characters form a number that is less than "26" and don't contain 0, they are treated the same as in case II.

V Combine cases

Knowing this, the full recurrence relation is:

$$F(n) = \begin{cases} F(n-1) + F(n-2) & \text{separate + combined} \\ F(n-2) & \text{only combine} \\ F(n-1) & \text{only separate} \\ 0 & \exists \geq 2 \text{ consecutive invalid chars} \end{cases}$$

Tips

There were many cases in this problem, which made it hard to track my thought process. I should try to look for different test cases on my own to find cases to implement in my code on my own rather than relying on submissions to determine the test cases needed.